Physics-Guided Attack Generation for Resilient Cyber-Physical State Estimation

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***Abstract*—This paper develops a resilient state-estimation framework for linear cyber-physical systems subject to sparse false-data injection attacks. A steady-state Kalman filter and a chi-square innovation detector are incorporated into a physics-guided attack generator. The generator uses the closed-loop attack-to-output map to quantify attack effectiveness and the attack-to-innovation map to quantify detection probability. Generated attacks are subsequently used to train a channel-wise localization network. Under explicit conditional-calibration and independence assumptions, calibrated safety probabilities are employed to select a measurement subset with a prescribed probability of being attack-free. The resulting analysis relates four quantities that are commonly treated separately: attack sparsity, sensor-set observability, localization reliability, and conditional Kalman-filter error covariance. A water-distribution case study reports the empirical observability profile and illustrates the localization improvement obtained from an earlier physics-guided training set; numerical validation of the revised KF/chi-square generator is left for future work.**

***Index Terms*—Resilient estimation, Kalman filter, chi-square detector, false data injection attack, generative model, multilayer perceptron, pruning, observability**

I. Introduction

Cyber-physical critical infrastructure relies on networked sensing, estimation, and control. This dependence creates an attack surface through which corrupted measurements can alter both the estimated state and the resulting control action. False-data injection attacks (FDIAs) have been studied extensively in power, water, and other infrastructure systems [1]–[4]. A particularly important class of FDIAs exploits knowledge of the estimator or detector to produce a significant physical effect while maintaining a small detection statistic [5].

Model-based bad-data detection commonly evaluates an innovation or residual against a prescribed threshold. Examples include generalized likelihood-ratio tests, Bayesian detectors, and chi-square tests associated with Kalman filtering [5]–[7]. These methods offer interpretable statistical thresholds, but their performance depends on the plant and noise models. Data-driven classifiers provide a complementary mechanism for identifying compromised channels and have been applied to FDIA detection and localization [8]–[10]. Their effectiveness, however, depends strongly on whether the training data represent attacks near the decision boundary.

Adversarial data generation addresses this limitation by enriching the training distribution. Existing approaches include optimization-based attack synthesis and generative models trained from attack corpora [11]–[13]. The former can be computationally demanding, whereas the latter presuppose a representative collection of feasible attacks. A physics-guided generator provides an alternative: known plant and estimator dynamics can be used as fixed discriminators, allowing attack samples to be generated without first collecting a labeled attack dataset.

Attack localization alone does not ensure resilient estimation. A localization model may retain compromised measurements or discard measurements that are important for observability. Secure and resilient estimators therefore exploit attack sparsity, sensor redundancy, and support information [14]–[17]. Prior work has investigated ℓ_1 estimators, support-weighted estimators, and resilient Kalman filters [18], [19]. A remaining design question is how localization reliability should be translated into a sensor subset whose cleanliness, observability, and estimation error can be quantified simultaneously.

Contributions: This paper addresses that question in a Kalman-filter setting. First, finite-horizon maps are derived from a sparse measurement attack to the closed-loop critical output and to the normalized Kalman-filter innovation. These maps define physics-based discriminators for generating attacks subject to an explicit chi-square detection-probability constraint. Second, a channel-wise localization network is combined with a calibrated sensor-selection rule. Under stated conditional-calibration and independence assumptions, the rule guarantees that the particular subset supplied to the estimator is attack-free with a prescribed probability. Third, the analysis relates the realized attack sparsity, the observability threshold, the selection reliability, and the conditional subset-KF covariance. This relation quantifies the estimation penalty incurred when higher selection reliability requires fewer measurements.

The remainder of the paper is organized as follows. Section II introduces the notation. Section III presents the

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stochastic CPS model, Kalman filter, and chi-square detector. Section IV derives the attack-response maps and formulates physics-guided attack generation. Section V develops channel-wise localization and calibrated sensor selection. Section VI analyzes the resulting subset Kalman filter. Section VII presents the water-distribution case study, and Section VIII concludes the paper.

II. Notation and Preliminary

The sets of real numbers, real n-vectors, and real $m \times n$ matrices are denoted by $\mathbb{R}$, $\mathbb{R}^n$, and $\mathbb{R}^{m\times n}$, respectively. For a matrix X, $X^\top$, $X^\dagger$, $\rho(X)$, and $\mathrm{tr}(X)$ denote its transpose, Moore–Penrose inverse, spectral radius, and trace. The identity matrix of dimension n is denoted by I_n. For symmetric matrices, $X \succeq 0$ and $X \succ 0$ denote positive semidefiniteness and positive definiteness.

For an index set $\mathcal{S}$, $X_{\mathcal{S}}$ denotes the submatrix formed from the rows of X indexed by $\mathcal{S}$, and $\mathcal{S}^c$ denotes its complement relative to the applicable sensor set. The cardinality of $\mathcal{S}$ is $|\mathcal{S}|$. The support and zero norm of a vector are

$$\mathrm{supp}(\mathbf{x}) = \{j : x_j \neq 0\}, \qquad \|\mathbf{x}\|_0 = |\,\mathrm{supp}(\mathbf{x})|.$$

A vector is k-sparse if $\|\mathbf{x}\|_0 \leq k$. A time window containing T samples and ending at index i is denoted by $\mathcal{I} = [i-T+1, i]$. Stacked quantities over this window carry the subscript $\mathcal{I}$.

The expectation and probability operators are denoted by $\mathbb{E}$ and Pr, respectively. The Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance P is $\mathcal{N}(\boldsymbol{\mu}, P)$. The central and noncentral chi-square distributions with ν degrees of freedom and noncentrality parameter λ are denoted by χ^2_ν and $\chi'^2_\nu(\lambda)$. Their cumulative distribution functions are denoted by $F_{\chi^2_\nu}$ and $F_{\chi'^2_\nu(\lambda)}$.

III. System and Threat Model

Consider the discrete-time stochastic linear system

$$\begin{aligned} \mathbf{x}_{i+1} &= A\mathbf{x}_i + B\mathbf{u}_i + \mathbf{w}_i, \\ \mathbf{y}_i &= C\mathbf{x}_i + \mathbf{v}_i, \end{aligned} \tag{1}$$

where $\mathbf{x}_i \in \mathbb{R}^n$, $\mathbf{u}_i \in \mathbb{R}^p$, and $\mathbf{y}_i \in \mathbb{R}^m$ denote the state, control input, and measurement, respectively. The process noise $\mathbf{w}_i \sim \mathcal{N}(\mathbf{0}, Q)$ and measurement noise $\mathbf{v}_i \sim \mathcal{N}(\mathbf{0}, R)$ are mutually independent white sequences, with $Q \succeq 0$ and $R \succ 0$. The initial state is Gaussian and independent of both noise sequences. The following assumptions are imposed:

- A1. The pair (A, B) is stabilizable.
- A2. There exists a positive integer $k_0 \leq m$ such that the pair $(A, C_{\mathcal{S}})$ is observable for all $\mathcal{S} \subset \{1, 2, \cdots, m\}$ with $|\mathcal{S}| \geq k_0$.
- A3. The pair $(A, Q^{1/2})$ is stabilizable.

Assumption A2 quantifies the redundancy of the sensing architecture. Specifically, loss of as many as $m - k_0$ sensors does not destroy observability, provided that the remaining measurements are attack-free.

A. Kalman filter

Let $\hat{\mathbf{x}}_{i|i-1}$ and $P_{i|i-1}$ denote the one-step state prediction and its error covariance. With a correctly initialized covariance, the Kalman filter recursion is

$$\hat{\mathbf{x}}_{i|i-1} = A\hat{\mathbf{x}}_{i-1|i-1} + B\mathbf{u}_{i-1}, \tag{2}$$
$$P_{i|i-1} = AP_{i-1|i-1}A^\top + Q, \tag{3}$$
$$\mathbf{r}_i = \mathbf{y}_i - C\hat{\mathbf{x}}_{i|i-1}, \tag{4}$$
$$S_i = CP_{i|i-1}C^\top + R, \tag{5}$$
$$K_i = P_{i|i-1}C^\top S_i^{-1}, \tag{6}$$
$$\hat{\mathbf{x}}_{i|i} = \hat{\mathbf{x}}_{i|i-1} + K_i\mathbf{r}_i, \tag{7}$$
$$P_{i|i} = (I - K_iC)\,P_{i|i-1}. \tag{8}$$

Under the nominal hypothesis, the innovation sequence is independent and satisfies $\mathbf{r}_i \sim \mathcal{N}(\mathbf{0}, S_i)$. Under A2–A3, the covariance recursion converges for every measurement subset covered by A2. In the full-sensor case, denote the limiting predicted covariance, gain, and innovation covariance by P^-, K, and $S = CP^-C^\top + R$, respectively. Hereafter, $\hat{\mathbf{x}}_i \triangleq \hat{\mathbf{x}}_{i|i}$.

B. Chi-square bad-data detector

The normalized innovation squared is

$$g_i \triangleq \mathbf{r}_i^\top S_i^{-1}\mathbf{r}_i \tag{9}$$

Under the nominal hypothesis $\mathcal{H}_0$, $g_i \sim \chi^2_m$. For a prescribed nominal false-alarm probability $\beta \in (0, 1)$, define

$$\mathsf{BDD}(\mathbf{y}_i) = \begin{cases} 0, & \text{if } g_i \leq \tau_{\chi^2}(\beta), \\ 1, & \text{otherwise,} \end{cases} \tag{10}$$

where $\tau_{\chi^2}(\beta) = F^{-1}_{\chi^2_m}(1-\beta)$. For a window $I = [i-T+1, i]$, independence of the nominal innovations gives

$$g_I \triangleq \sum_{j\in I} g_j \sim \chi^2_{Tm}. \tag{11}$$

Thus the detector threshold has a direct nominal false-alarm interpretation. Its probability of detection under attack depends on the attack-induced noncentrality parameter developed in Section IV.

C. Closed-loop control

The state estimate is supplied to the certainty-equivalence control law

$$\mathbf{u}_i = K_p\hat{\mathbf{x}}_i, \tag{12}$$

where K_p is selected such that $\rho(A+BK_p) < 1$. Substitution into (1) gives

$$\mathbf{x}_{i+1} = (A + BK_p)\,\mathbf{x}_i + BK_p\,(\hat{\mathbf{x}}_i - \mathbf{x}_i) + \mathbf{w}_i.$$

Consequently, bounded second moments of the estimation error imply bounded second moments of the closed-loop state. In the attack-free case, $\mathbb{E}\|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2^2 = \mathrm{tr}(P_{i|i})$, and the mean closed-loop dynamics are asymptotically stable [20], [21].

IV. Automated Attack Generation

Consider the attacked measurement model

$$\mathbf{y}_i^a = C\mathbf{x}_i^a + \mathbf{v}_i + \mathbf{e}_i, \tag{13}$$

where $\mathbf{e}_i \in \mathbb{R}^m$ is k-sparse. Superscripts a and 0 denote attacked and nominal closed-loop trajectories driven by identical realizations of the process and measurement noises. The analysis considers deterministic attacks that are independent of these noises and begin at the left endpoint of the detection window. The objective of the attack generator is to produce sequences that significantly perturb a critical output while remaining difficult to detect using (10).

A. Attack-induced innovation and output maps

The discriminator construction is developed for the steady-state Kalman gain K. Define

$$\delta\mathbf{x}_i = \mathbf{x}_i^a - \mathbf{x}_i^0, \qquad \delta\hat{\mathbf{x}}_{i|i-1} = \hat{\mathbf{x}}_{i|i-1}^a - \hat{\mathbf{x}}_{i|i-1}^0,$$

and let $\boldsymbol{\zeta}_i = \mathrm{col}(\delta\mathbf{x}_i, \delta\hat{\mathbf{x}}_{i|i-1})$. The attack-induced innovation is

$$\delta\mathbf{r}_i = \mathbf{r}_i^a - \mathbf{r}_i^0 = D\boldsymbol{\zeta}_i + \mathbf{e}_i, \qquad D = \begin{bmatrix} C & -C \end{bmatrix}. \tag{14}$$

Under the control law (12), direct substitution of the KF update gives the augmented recursion

$$\boldsymbol{\zeta}_{i+1} = F_a\boldsymbol{\zeta}_i + G_a\mathbf{e}_i, \tag{15}$$

where, with $A_c = A + BK_p$,

$$F_a = \begin{bmatrix} A + BK_pKC & BK_p(I - KC) \\ A_cKC & A_c(I - KC) \end{bmatrix}, \quad G_a = \begin{bmatrix} BK_pK \\ A_cK \end{bmatrix}. \tag{16}$$

Equations (14)–(16) account for the propagation of earlier attacks through both the plant and the estimator. For a window $\mathcal{I} = [i - T + 1, i]$ and zero attack-induced deviation at its beginning, unrolling these equations yields

$$\delta\mathbf{r}_\mathcal{I} = \mathcal{H}_r\mathbf{e}_\mathcal{I}, \qquad M_2 = \mathrm{blkdiag}(S^{-1/2}, \ldots, S^{-1/2})\mathcal{H}_r, \tag{17}$$

where $\mathcal{H}_r$ is block lower triangular, with identity diagonal blocks and subdiagonal block (p, q) equal to $DF_a^{p-q-1}G_a$ for $p > q$.

Let $\mathbf{y}_{c,i} = C_c\mathbf{x}_i$ denote the critical output. The corresponding stacked perturbation satisfies

$$\delta\mathbf{y}_{c,\mathcal{I}} = \mathcal{H}_c\mathbf{e}_\mathcal{I} \triangleq M_1\mathbf{e}_\mathcal{I}, \tag{18}$$

where $\mathcal{H}_c$ is obtained by unrolling (15) with output matrix $\begin{bmatrix} C_c & 0 \end{bmatrix}$. Thus M_1 captures the complete plant–controller–estimator response, whereas M_2 captures the complete normalized innovation response.

B. Successful attacks and generative model

Under the deterministic, noise-independent, window-initialized attack model stated above, the windowed statistic has the noncentral chi-square distribution

$$g_\mathcal{I}^a \sim \chi'^2_{Tm}(\lambda_\mathcal{I}), \qquad \lambda_\mathcal{I} = \|M_2\mathbf{e}_\mathcal{I}\|_2^2. \tag{19}$$

Accordingly, $\lambda_\mathcal{I}$ determines the detection probability

$$P_D(\mathbf{e}_\mathcal{I}) = 1 - F_{\chi'^2_{Tm}(\lambda_\mathcal{I})}\big(\tau_{\chi^2,\mathcal{I}}(\beta)\big), \tag{20}$$

where $\tau_{\chi^2,\mathcal{I}}(\beta) = F^{-1}_{\chi^2_{Tm}}(1 - \beta)$.

Definition 1 ((α,γ)-successful FDIA)**.** *Define the per-time sparsity set*

$$\Sigma_k^{(T)} = \{\mathbf{e}_\mathcal{I} : \|\mathbf{e}_j\|_0 \le k,\ j \in \mathcal{I}\}.$$

An attack sequence $\mathbf{e}_\mathcal{I} \in \Sigma_k^{(T)}$ is (α,γ)-successful if

$$\|M_1\mathbf{e}_\mathcal{I}\|_2 \ge \alpha, \qquad P_D(\mathbf{e}_\mathcal{I}) \le \gamma, \tag{21}$$

where $\alpha > 0$ is the required critical-output perturbation and $\gamma \in (\beta, 1)$ is the maximum admissible detection probability.

Because (20) is monotone in $\lambda_\mathcal{I}$, the second condition is equivalently represented by

$$\|M_2\mathbf{e}_\mathcal{I}\|_2^2 \le \lambda_{\max}(\beta,\gamma), \tag{22}$$

where $\lambda_{\max}$ is the unique noncentrality level that yields $P_D = \gamma$. This replaces an ad hoc residual threshold with a detector-calibrated stealthiness constraint.

The feasible attack set is therefore

$$\mathcal{S}(\alpha,\gamma) = \{\mathbf{e} \in \Sigma_k^{(T)} : \|M_1\mathbf{e}\|_2 \ge \alpha, \ \|M_2\mathbf{e}\|_2^2 \le \lambda_{\max}(\beta,\gamma)\}. \tag{23}$$

We approximate this set using a neural generator $G(\mathbf{z};\boldsymbol{\theta}_g)$, with $\mathbf{z} \sim P_z$. Let $\Pi_{k,T}$ retain the k largest-magnitude entries in each time block and set all other entries to zero. The generated attack is defined as

$$\mathbf{e}_\mathcal{I} = \Pi_{k,T}(G(\mathbf{z};\boldsymbol{\theta}_g)), \tag{24}$$

which enforces membership in $\Sigma_k^{(T)}$. The plant and filter maps M_1 and M_2 act as fixed physics-based discriminators, as illustrated in Fig. 1.

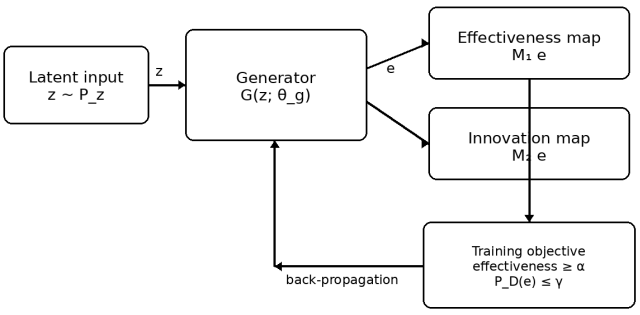


Fig. 1. Physics-guided attack generation. The discriminator M_1 evaluates critical-output effectiveness, whereas M_2 determines the noncentrality parameter of the windowed chi-square detector.

The generator is trained using

$$J(\mathbf{e}) = \mathrm{ReLU}(\alpha - \|M_1\mathbf{e}\|_2) + \mathrm{ReLU}\big(\|M_2\mathbf{e}\|_2^2 - \lambda_{\max}(\beta,\gamma)\big), \quad (25)$$

$$\boldsymbol{\theta}_g^\star = \arg\min_{\boldsymbol{\theta}_g} \mathbb{E}_{\mathbf{z}\sim P_z}\left[J(\Pi_{k,T}(G(\mathbf{z};\boldsymbol{\theta}_g)))\right]. \quad (26)$$

The resulting labeled samples are used to train the channel-wise localization model in Section V.

V. Detection and Localization

The attacks generated in Section IV are used to train a channel-wise classifier. The classifier serves two distinct purposes. First, it identifies measurement channels that are likely to be compromised. Second, its calibrated channel probabilities are used to construct a sensor subset for resilient state estimation.

A. Channel-wise localization

Let $\mathcal{T} \subseteq \{1,\dots,m\}$ denote the attacked sensor set and define the safe-sensor indicator

$$q_j = \begin{cases} 0, & j \in \mathcal{T}, \\ 1, & j \notin \mathcal{T}. \end{cases} \quad (27)$$

The localization model is a vector-valued map $g(\mathbf{y};\boldsymbol{\theta}) \in (0,1)^m$, whose jth component estimates the probability that sensor j is safe. In the numerical study, g is a four-layer fully connected network with three ReLU hidden layers and a sigmoid output layer. Its binary decision is

$$\hat{q}_j = \mathbb{I}\big\{g_j(\mathbf{y};\boldsymbol{\theta}) \geq \tfrac{1}{2}\big\}. \quad (28)$$

Given training pairs $\{(\mathbf{y}^{(i)},\mathbf{q}^{(i)})\}_{i=1}^N$, the parameters are obtained by minimizing the channel-wise binary cross-entropy

$$J(\boldsymbol{\theta}) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{m}\big[q_j^{(i)}\log g_j(\mathbf{y}^{(i)};\boldsymbol{\theta}) + (1-q_j^{(i)})\log(1-g_j(\mathbf{y}^{(i)};\boldsymbol{\theta}))\big]. \quad (29)$$

The generative model enriches the training set with attacks near the effectiveness and detectability boundaries in (21).

B. Calibrated safe-set pruning

Let $\hat{\mathcal{T}}^c = \{j : \hat{q}_j = 1\}$ be the set declared safe by the classifier, and let $\mathcal{G}$ denote the sigma-field generated by the complete classifier output and the resulting ordering information. A held-out calibration set is used to estimate, for each candidate $j \in \hat{\mathcal{T}}^c$, the conditional safety probability

$$p_j = \Pr\{q_j = 1 \mid \mathcal{G}\}, \quad (30)$$

where calibration may be implemented by conditioning on score bins and other features included in $\mathcal{G}$. The following analysis assumes that these conditional probabilities are calibrated and that the channel safety events are conditionally independent given $\mathcal{G}$. These assumptions can be relaxed by replacing the product below with a joint lower confidence bound.

Order the declared-safe indices so that $p_{j_1} \geq p_{j_2} \geq \cdots \geq p_{j_{|\hat{\mathcal{T}}^c|}}$. For a prescribed reliability $\eta \in (0,1)$, define

$$l_\eta = \max\left\{l \in \{0,\dots,|\hat{\mathcal{T}}^c|\} : \prod_{r=1}^{l} p_{j_r} \geq \eta\right\}, \quad (31)$$

with the empty product equal to one, and select

$$\hat{\mathcal{T}}_\eta^c = \{j_1,\dots,j_{l_\eta}\}. \quad (32)$$

This construction differs from a count-based confidence bound: it certifies the particular prefix supplied to the estimator.

Proposition 1 (Safe-set precision)**.** *Under the calibration and conditional-independence assumptions,*

$$\Pr\{\hat{\mathcal{T}}_\eta^c \cap \mathcal{T} = \varnothing \mid \mathcal{G}\} = \prod_{r=1}^{l_\eta} p_{j_r} \geq \eta. \quad (33)$$

Equivalently, the safe-set positive predictive value

$$\mathrm{PPV}_{\mathrm{safe}} = \frac{|\hat{\mathcal{T}}_\eta^c \cap \mathcal{T}^c|}{|\hat{\mathcal{T}}_\eta^c|} \quad (34)$$

equals one with probability at least η whenever $l_\eta > 0$.

Proof. The selected set is clean precisely when all l_η selected channels are safe. Conditional independence given $\mathcal{G}$ and (30) give the product in (33); the inequality follows from (31). Taking expectations with respect to $\mathcal{G}$ also gives the unconditional bound $\Pr\{\hat{\mathcal{T}}_\eta^c \cap \mathcal{T} = \varnothing\} \geq \eta$. □

C. Sparsity–observability–precision relation

The next result identifies the admissible reliability levels for which pruning preserves observability.

Proposition 2 (Precision and observability)**.** *Suppose $\hat{\mathcal{T}}^c$ contains at least k_0 candidates. The pruned set is observable under A2 and is clean with probability at least η if*

$$\eta \leq \eta_{\mathrm{obs}} \triangleq \prod_{r=1}^{k_0} p_{j_r}. \quad (35)$$

Moreover, l_η is nonincreasing in η.

Proof. Condition (35) makes $l = k_0$ feasible in (31); hence $l_\eta \geq k_0$, and A2 implies observability. Cleanliness follows from Proposition 1. Finally, increasing η can only remove feasible values from the maximization in (31). □

Attack sparsity imposes a separate physical limit. Let $k = |\mathcal{T}|$ denote the realized number of attacked sensors. Exactly $m-k$ sensors are genuinely safe; hence, the existence of a clean observable subset requires

$$m - k \geq k_0, \qquad \text{or equivalently} \qquad k \leq m - k_0. \quad (36)$$

False-safe declarations cannot relax (36); they only reduce the probability that the selected set is clean. Thus, (36) quantifies the sparsity–observability limit, while (35) quantifies the precision required to realize that limit using a learned localization model.

Remark 1. *The chi-square detector and the localization network quantify different events. Equation* (20) *describes the probability of a global alarm, whereas* (33) *describes the probability that the particular sensor subset supplied to the estimator is uncontaminated. Both quantities enter the resilient-estimation analysis, but they should not be conflated.*

Algorithm 1: Attack localization and calibrated safe-set pruning

Offline:

1) Construct M_1 and M_2 from (18)–(17).
2) Generate labeled attacks by solving (26).
3) Train the channel-wise classifier using (29).
4) Calibrate the conditional safety probabilities in (30).

Online:

1) Evaluate $g(\mathbf{y};\boldsymbol{\theta})$ and form $\hat{\mathcal{T}}^c$.
2) Sort the candidates by calibrated safety probability.
3) Compute l_η from (31) and return $\hat{\mathcal{T}}^c_\eta$ from (32).

VI. Resilient Estimator

The pruned set $\hat{\mathcal{T}}^c_\eta$ is used to define a reduced-order measurement update. To preserve the nominal chi-square calibration, the set employed at time i is determined from measurements available no later than time $i-1$. The asymptotic analysis below is restricted to a subset $\mathcal{S}$ that remains fixed over the estimation horizon. Time-varying selection requires a separate analysis of the switched Riccati recursion.

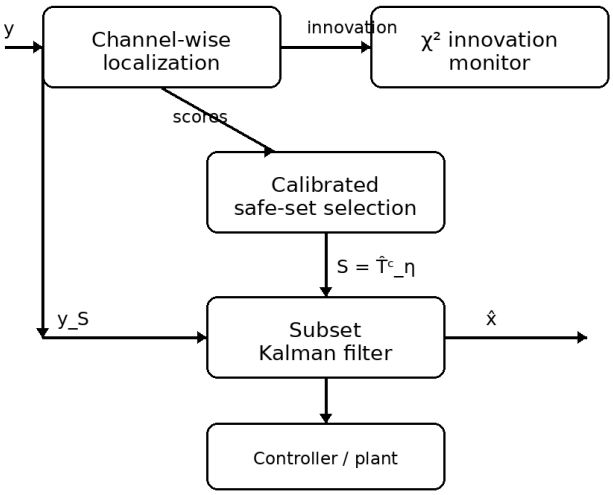


Fig. 2. Resilient estimation architecture. The localization and pruning stages select a sensor subset, and both the Kalman update and chi-square test are formed using that subset.

A. Subset Kalman filter

For a selected set $\mathcal{S} \subseteq \{1,\dots,m\}$, define

$$\mathbf{y}_{\mathcal{S},i} = C_{\mathcal{S}}\mathbf{x}_i + \mathbf{v}_{\mathcal{S},i} + \mathbf{e}_{\mathcal{S},i}. \tag{37}$$

The resilient filter applies (2)–(8) with (C,R) replaced by $(C_{\mathcal{S}}, R_{\mathcal{S}})$. Its innovation statistic is

$$g_i^{\mathcal{S}} = (\mathbf{r}_i^{\mathcal{S}})^\top (S_i^{\mathcal{S}})^{-1}\mathbf{r}_i^{\mathcal{S}}. \tag{38}$$

When $\mathcal{S}$ is selected independently of the current innovation and is attack-free, $g_i^{\mathcal{S}} \sim \chi^2_{|\mathcal{S}|}$. Hence, the threshold must be recomputed using $|\mathcal{S}|$ degrees of freedom; a window of T independent subset innovations has $T|\mathcal{S}|$ degrees of freedom.

For a fixed observable subset $\mathcal{S}$, let $P_\infty(\mathcal{S})$ denote the stabilizing filtered solution of the corresponding Riccati equation. Since $Q \succeq 0$, this covariance is positive semidefinite. Additional independent measurements cannot increase the optimal filtering covariance. Therefore, for nested observable sets $\mathcal{S}_1 \subseteq \mathcal{S}_2$,

$$P_\infty(\mathcal{S}_2) \preceq P_\infty(\mathcal{S}_1). \tag{39}$$

Theorem VI.1 (Conditional resilient-estimation performance)**.** *Let $\mathcal{S} = \hat{\mathcal{T}}^c_\eta$ be fixed over the estimation horizon and suppose $\eta \le \eta_{\mathrm{obs}}$ in* (35)*. Conditional on the event*

$$\mathcal{E}_{\mathcal{S}} = \{\mathcal{S} \cap \mathcal{T} = \varnothing\},$$

the resilient filter is identical to the attack-free Kalman filter based on $\mathcal{S}$, and

$$\lim_{i\to\infty} \mathbb{E}\Big[\|\mathbf{x}_i - \hat{\mathbf{x}}^{\mathcal{S}}_{i|i}\|_2^2 \mid \mathcal{E}_{\mathcal{S}}, \mathcal{S}\Big] = \operatorname{tr} P_\infty(\mathcal{S}). \tag{40}$$

Furthermore,

$$\begin{aligned}\operatorname{tr} P_\infty(\{1,\dots,m\}) &\le \operatorname{tr} P_\infty(\mathcal{S}) \\ &\le \max_{\mathcal{U}\subseteq\mathcal{S},\ |\mathcal{U}|=k_0} \operatorname{tr} P_\infty(\mathcal{U}).\end{aligned} \tag{41}$$

The conditioning event satisfies $\Pr(\mathcal{E}_{\mathcal{S}} \mid \mathcal{G}) \ge \eta$ and, consequently, $\Pr(\mathcal{E}_{\mathcal{S}}) \ge \eta$.

Proof. Propositions 1 and 2 give $\Pr(\mathcal{E}_{\mathcal{S}} \mid \mathcal{G}) \ge \eta$ and $|\mathcal{S}| \ge k_0$. Assumption A2 then implies that $(A, C_{\mathcal{S}})$ is observable. On $\mathcal{E}_{\mathcal{S}}$, $\mathbf{e}_{\mathcal{S},i} = 0$ throughout the interval, so the subset-filter recursion coincides exactly with its attack-free counterpart. Standard Kalman-filter convergence gives (40). The first inequality in (41) follows because $\mathcal{S}$ is contained in the full sensor set. For every k_0-element subset $\mathcal{U} \subseteq \mathcal{S}$, (39) gives $P_\infty(\mathcal{S}) \preceq P_\infty(\mathcal{U})$; taking the largest trace proves the second inequality. □

The theorem is intentionally conditional. No finite unconditional mean-square bound follows from a cleanliness probability alone because the attack magnitude on the complementary event $\mathcal{E}^c_{\mathcal{S}}$ may be unbounded. A trajectory-wide unconditional guarantee therefore requires either an attack-amplitude model or a robust update on the failure event.

B. Quantitative performance relation

Equations (36), (35), and (41) provide the desired quantitative relation. First, $k \le m - k_0$ is necessary for any clean observable subset to exist. Second, the calibrated probabilities determine the largest reliability level η_{obs} for which the learned selector retains at least k_0 sensors. Third, for every admissible η, the resulting conditional mean-square error is $\operatorname{tr} P_\infty(\mathcal{S})$.

The relation also exposes a design tradeoff. Increasing η generally reduces l_η, thereby increasing the covariance through (39); decreasing η retains more measurements but weakens the certified probability that all retained sensors are safe. A suitable reliability level may therefore be selected by

$$\min_{\eta\in[\eta_{\min},\eta_{\text{obs}}]} \operatorname{tr} P_\infty(\hat{\mathcal{T}}_\eta^c) \quad \text{subject to} \quad \Pr(\mathcal{E}_{\hat{\mathcal{T}}_\eta^c} \mid \mathcal{G}) \geq \eta. \tag{42}$$

Here $\eta_{\min}$ is a designer-specified reliability floor. This formulation connects localization confidence directly to the estimator covariance rather than treating observability as a binary post-processing check.

VII. Simulation

This section evaluates the sensing redundancy and channel-localization components of the proposed framework using a water-distribution CPS. The available data establish the empirical observability profile and quantify the effect of physics-guided attack data on localization. The scope of the study is limited to these two components; no KF/χ^2 performance claim is inferred from residual-based simulations of the earlier estimator.

A. Water-distribution model

The system in Fig. 3 contains ten operating tanks and one storage tank. The control objective is to regulate the operating-tank levels by adjusting the inlet valves. Random consumer demand is modeled as a process disturbance. Level and differential-pressure measurements are transmitted to the control center and may be compromised by additive FDIAs.

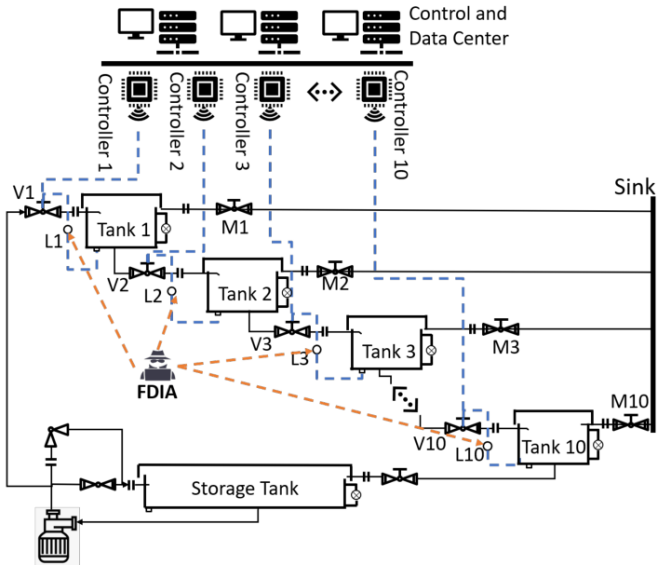


Fig. 3. Water-distribution CPS. Solid lines denote water pipelines, blue dotted lines denote measurement and control communication, and orange dotted lines denote attack routes.

The linearized discrete-time model is

$$\begin{aligned} \mathbf{x}_{i+1} &= A\mathbf{x}_i + B\mathbf{u}_i - \mathbf{d}_i, \\ \mathbf{y}_i &= C\mathbf{x}_i + \mathbf{v}_i + \mathbf{e}_i, \end{aligned} \tag{43}$$

where $\mathbf{x}, \mathbf{u} \in \mathbb{R}^{10}$, $\mathbf{d} \in \mathbb{R}^{10}$ is the demand disturbance, and $\mathbf{e}$ is the measurement attack. The critical output is the aggregate level of the operating tanks, i.e., $C_c = \mathbf{1}_{10}^\top$. Model-generation and attack-generation code is available at https://github.com/ZYblend/Physics-guided_GAN_Attack_Generation.

B. Empirical observability profile

For each sensor-set cardinality, 50,000 subsets were sampled uniformly and the rank of the associated observability matrix was evaluated. Figure 4 reports the fraction of sampled subsets for which (A, C_S) was observable.

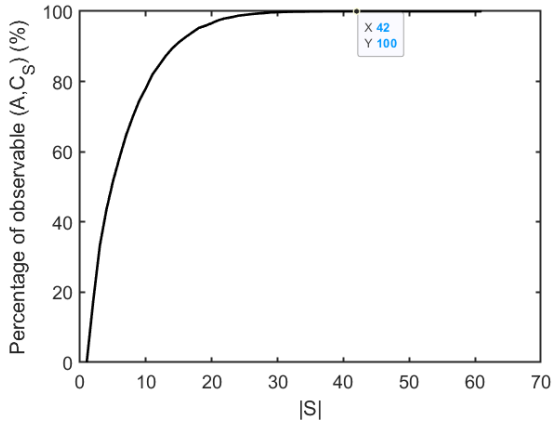


Fig. 4. Empirical frequency of observability as a function of the number of retained measurements (50,000 random subsets per cardinality).

All sampled subsets containing at least 42 measurements were observable. This observation motivates the empirical design value $k_0 = 42$; it does not constitute an exhaustive verification of A2. With $m = 61$, the corresponding empirical sparsity guideline is $k \leq 19$. Experiments with more than 19 attacked channels lie outside the sampled operating regime.

C. Localization performance

Two four-layer MLPs were evaluated. MLP1 was trained using random additive attacks; MLP2 used the same data augmented with attacks produced by the earlier residual-based physics-guided generator. Thus, this experiment evaluates the general benefit of physics-guided data augmentation, rather than the revised generator in Section IV. Both networks contain three hidden layers with 512, 1024, and 512 ReLU units, followed by an m-dimensional sigmoid layer. The comparison uses a common test set of 50,000 FDIAs.

The mean localization precision increased from 74.37% for MLP1 to 98.69% for MLP2 on the reported test distribution. Together with the empirical value $k_0 = 42$, calibrated classifier scores would determine the admissible reliability η_{obs} through (35). The archived experiment did not record the conditional calibration data or subset Riccati covariances required to evaluate this relation; consequently, no clean-set probability or KF covariance certificate is claimed for these results.

VIII. Conclusion

This paper developed a resilient estimation architecture that combines a Kalman filter, chi-square innovation monitoring,

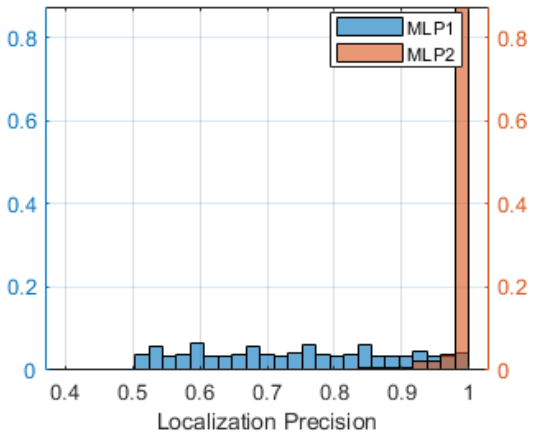


Fig. 5. Distribution of channel-localization precision on a common test set. MLP1 is trained using random attacks; MLP2 is trained using random and physics-guided attacks.

physics-guided attack generation, and channel-wise localization. The attack generator was formulated using finite-horizon maps that retain the dynamics of both the estimator and the controlled plant. This construction gives a detector-calibrated measure of stealthiness through the noncentral chi-square distribution, rather than through an unnormalized residual threshold.

The localization output was converted into an estimator input through calibrated safety probabilities. Under explicit calibration and conditional-independence assumptions, the selected subset is attack-free with a prescribed probability. The analysis further showed that attack sparsity limits the existence of a clean observable subset, while the size of the selected subset determines the conditional Kalman-filter covariance through the Riccati equation. The water-distribution case study demonstrated the empirical sensing redundancy and the localization improvement obtained by augmenting the training set with physics-guided attacks. Closed-loop numerical validation of the revised KF/χ^2 architecture is outside the scope of the available data and is not inferred from the earlier residual-based results.

Future work will consider dependent channel-calibration models, time-varying sensor subsets, and unconditional error guarantees obtained by combining probabilistic selection with a robust update on the selection-failure event.

REFERENCES

[1] Y. Liu, P. Ning, and M. K. Reiter, "False data injection attacks against state estimation in electric power grids," *ACM Transactions on Information and System Security (TISSEC)*, vol. 14, no. 1, p. 13, 2011.

[2] Liang *et al.*, "The 2015 ukraine blackout: Implications for false data injection attacks," *IEEE Transactions on Power Systems*, vol. 32, no. 4, pp. 3317–3318, 2016.

[3] R. M. Clark, S. Panguluri, T. D. Nelson, and R. P. Wyman, "Protecting drinking water utilities from cyberthreats," *Journal of the American Water Works Association*, vol. 109, no. INL/JOU-16-39302, 2017.

[4] B. Brentan, P. Rezende, D. Barros, G. Meirelles, E. Luvizotto, and J. Izquierdo, "Cyber-attack detection in water distribution systems based on blind sources separation technique," *Water*, vol. 13, no. 6, p. 795, 2021.

[5] Y. Mo and B. Sinopoli, "False data injection attacks in control systems," in *Preprints of the 1st workshop on Secure Control Systems*, 2010, pp. 1–6.

[6] B. Tang, J. Yan, S. Kay, and H. He, "Detection of false data injection attacks in smart grid under colored gaussian noise," in *2016 IEEE Conference on Communications and Network Security (CNS)*. IEEE, 2016, pp. 172–179.

[7] D. Ding, Q.-L. Han, Y. Xiang, X. Ge, and X.-M. Zhang, "A survey on security control and attack detection for industrial cyber-physical systems," *Neurocomputing*, vol. 275, pp. 1674–1683, 2018.

[8] M. Ozay *et al.*, "Machine learning methods for attack detection in the smart grid," *IEEE transactions on neural networks and learning systems*, vol. 27, no. 8, pp. 1773–1786, 2016.

[9] O. Boyaci, M. R. Narimani, K. Davis, M. Ismail, T. J. Overbye, and E. Serpedin, "Joint detection and localization of stealth false data injection attacks in smart grids using graph neural networks," *arXiv preprint arXiv:2104.11846*, 2021.

[10] Esmalifalak *et al.*, "Stealth false data injection using independent component analysis in smart grid," in *SmartGridComm, IEEE International Conference on*. IEEE, 2011, pp. 244–248.

[11] Q. Yang, J. Yang, W. Yu, D. An, N. Zhang, and W. Zhao, "On false data-injection attacks against power system state estimation: Modeling and countermeasures," *IEEE Transactions on Parallel and Distributed Systems*, vol. 25, no. 3, pp. 717–729, 2013.

[12] M. Mohammadpourfard, F. Ghanaatpishe, M. Mohammadi, S. Lakshminarayana, and M. Pechenizkiy, "Generation of false data injection attacks using conditional generative adversarial networks," in *2020 IEEE PES Innovative Smart Grid Technologies Europe (ISGT-Europe)*. IEEE, 2020, pp. 41–45.

[13] S. Ahmadian, H. Malki, and Z. Han, "Cyber attacks on smart energy grids using generative adverserial networks," in *2018 IEEE Global Conference on Signal and Information Processing (GlobalSIP)*. IEEE, 2018, pp. 942–946.

[14] H. Fawzi, P. Tabuada, and S. Diggavi, "Secure estimation and control for cyber-physical systems under adversarial attacks," *IEEE Transactions on Automatic control*, vol. 59, no. 6, pp. 1454–1467, 2014.

[15] M. Pajic, J. Weimer, N. Bezzo, P. Tabuada, O. Sokolsky, I. Lee, and G. J. Pappas, "Robustness of attack-resilient state estimators," in *2014 ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS)*. IEEE, 2014, pp. 163–174.

[16] M. Pajic, P. Tabuada, I. Lee, and G. J. Pappas, "Attack-resilient state estimation in the presence of noise," in *2015 54th IEEE Conference on Decision and Control (CDC)*. IEEE, 2015, pp. 5827–5832.

[17] Y. Zheng and O. M. Anubi, "Attack-resilient weighted ℓ_1 observer with prior pruning," in *2021 American control conference*. IEEE, 2021.

[18] C. Lee, H. Shim, and Y. Eun, "Secure and robust state estimation under sensor attacks, measurement noises, and process disturbances: Observer-based combinatorial approach," in *2015 European Control Conference (ECC)*. IEEE, 2015, pp. 1872–1877.

[19] Y. Zheng and O. M. Anubi, "Attack-resilient observer pruning for path-tracking control of wheeled mobile robot," in *2020 ASME Dynamic Systems and Control (DSC) Conference*. ASME, 2020, pp. 1–9.

[20] B. D. O. Anderson and J. B. Moore, *Optimal Filtering*. Dover Publications, 2012, reprint of 1979 Prentice-Hall edition.

[21] J. B. Rawlings, D. Q. Mayne, and M. Diehl, *Model predictive control: theory, computation, and design*. Nob Hill Publishing Madison, WI, 2017, vol. 2.